

Example Climate Change Impact/Risk Case Studies:

Within your groups, consider the following:

- What is the impact/risk you are assessing? Is the “impact” that you are assessing clear? Why or why not? If not, make an assumption in order to proceed
- What data do you need to make an assessment of the impact/risk?
- What additional data/information do you need in order to provide a complete assessment?
- Can you find this data/information through research or do you need to consult with other relevant community or sector participants?
- Distinguish between the climate data/information you need and any other information you may need to complete the assessment.

1. The Ontario Ministry of Labour is conducting a study to estimate the economic impacts of reduced productivity of construction workers as climate changes. Healthcare officials recommend the following guidelines for outdoor labour during heat events (note: humidex is a “feels like” temperature that takes humidity into account):

- Humidex reaches 34: provide extra water, decrease the workload.
- Humidex reaches 38: add an extra 15 minute break.
- Humidex reaches 42: work shifts need to be 30 minutes on, 30 minutes off.
- Humidex reaches 46: many types of labour will need to be restricted.

Your consulting firm has been contracted to provide a climate change impact assessment of the change in the frequency that each of these thresholds will be exceeded on varying time-horizons, year 2040, 2050 and 2060, for the Greater Toronto area of Ontario.

2. The Whistler-Blackcomb Ski Resort in British Columbia is in the process of a long-term strategic planning exercise and is discussing future investment in snow-making equipment. To inform their decision-making, they have contracted your consulting company to provide a climate change impact assessment of future temperature and snowfall changes, as well, as any changes in the start and end of the ski season for the time period 2040-2070.

3. G3 Canada Limited, formerly the Canadian Wheat Board, is examining emerging international markets for Canadian grain. In order to inform their research they are interested in projections of wheat and barley yield in the presence of climate change. They are interested in best and worst case scenarios for yield in the Canadian prairie region over the next 30 years.
4. The Ontario Ministry of Health has commissioned a study on the projected changes in the incidence of West Nile Virus infection in the presence of climate change. The Ontario Ministry of the Environment and Climate Change has been asked to provide a climate change impact assessment of the change in the number of days per year temperatures will be within the optimal development range of the mosquito for the entire province of Ontario.
5. The City of Toronto is examining how climate change will impact city beach closures due to contamination. One of the primary sources of beach contamination is from runoff from extreme precipitation. Your city department has been tasked with providing projections of the change in the number of days per summer that popular beach locations in Toronto will need to close for the years 2040 to 2070.